FitBuddy-AI Fitness Plan Generator Using Gemini Models

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Project Planning Phase

Here's the *Project Planning Phase* for FitBuddy – AI Fitness Plan Generator, structured into actionable steps and timelines:

🗓️ Project Planning Phase: FitBuddy

1. Project Scope

- *Core Deliverables*

- AI-powered workout plan generator (Gemini Pro).

- Nutrition/recovery tip generator (Gemini Flash).

- Feedback-based plan updater.

- Admin dashboard for trainers.

- Local deployment with FastAPI + Uvicorn.

- *Boundaries*

- Initial deployment limited to local environment.

- Cloud integration and wearable device support planned for later phases.

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2. Work Breakdown Structure (WBS)

- *Phase 1: Preparation*
  - Requirement gathering & analysis.
  - Environment setup (Python, FastAPI, SQLite, Gemini API).
- *Phase 2: Development*
  - Backend logic (routes, AI integration, DB).
  - Frontend templates (index, result, admin).
- *Phase 3: Testing*
  - Unit testing for AI functions.
  - Integration testing for DB + frontend.
- *Phase 4: Deployment*
  - Local deployment with Uvicorn.
  - Documentation & API testing.
- *Phase 5: Review*
  - User feedback collection.
  - Iterative improvements.

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3. Timeline (Milestones)

Milestone	Duration	Deliverable
Model Selection & Architecture	1 week	Gemini Pro & Flash integrated
Core Functionality Development	2 weeks	Workout, nutrition, feedback modules

| Routing Logic (routes.py) | 1 week | Functional API endpoints |

| Frontend Development | 2 weeks | Responsive UI templates |

| Deployment & Testing | 1 week | Local server running, verified |

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4. Resource Planning

- *Team Roles*
  - AI Engineer → Gemini integration.
  - Backend Developer → FastAPI + DB.
  - Frontend Developer → HTML/CSS/Jinja2.
  - QA Tester → Testing & validation.
  - Project Manager → Timeline & coordination.

- *Tools & Tech*
  - Python, FastAPI, SQLAlchemy, Jinja2.
  - Google Gemini API.
  - SQLite DB.
  - Uvicorn server.

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5. Risk Management

- *Technical Risks*
  - API downtime → Mitigation: fallback prompts or cached plans.
  - DB corruption → Mitigation: regular backups.
- *Operational Risks*
  - User adoption → Mitigation: intuitive UI + clear instructions.

- Scalability limits → Mitigation: plan for cloud deployment.

6. Success Criteria

- Users can generate and update personalized workout plans seamlessly.

- Nutrition tips are concise, relevant, and practical.

- Admins can monitor multiple users effectively.

- Application runs smoothly in local deployment with minimal errors.